

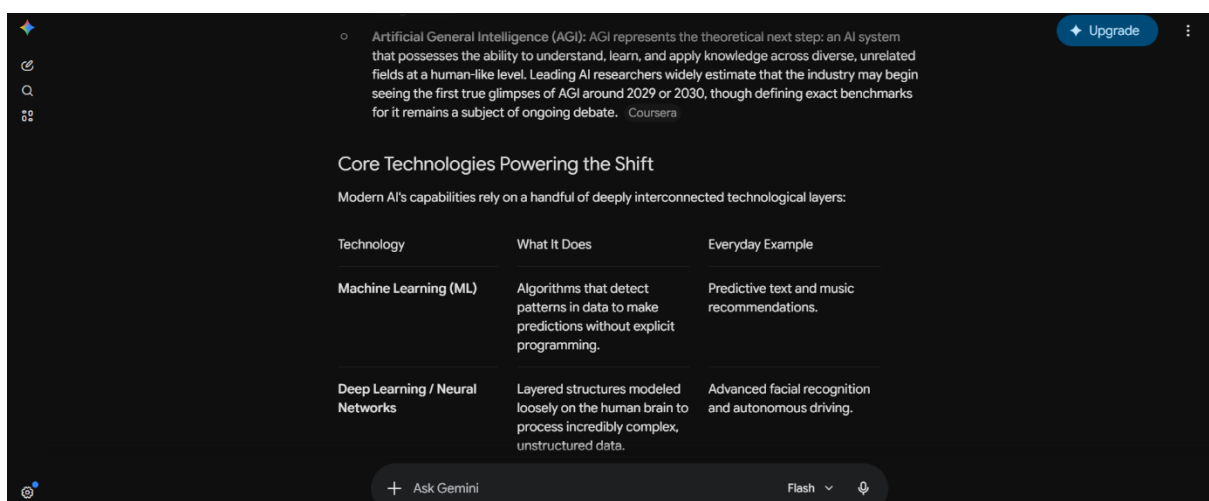
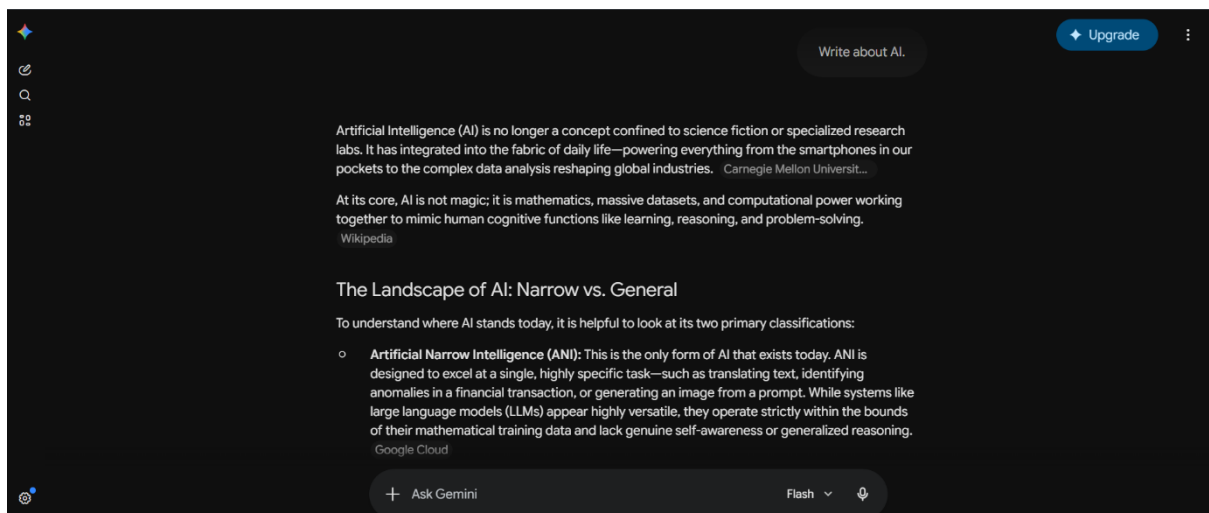
Task 01: Writing Better Prompts – Exploring Clarity, Specificity, and Tone in Prompt Engineering.

Objective:

To understand how prompt clarity, specificity, and tone influence AI-generated responses and to demonstrate how improved prompts produce more accurate, relevant, and useful outputs.

1. AI

Vague Prompt: Write about AI



Networks

loosely on the human brain to process incredibly complex, unstructured data.

and autonomous driving.

Natural Language Processing (NLP)

The branch of AI that allows machines to break down, interpret, and generate human language.

Customer service chatbots and real-time speech translation.

Generative AI

Deep learning models trained on vast datasets to create completely new content (text, audio, video, or code).

Creating custom graphics or summarizing long documents from a text prompt.

...

The Dual-Edged Sword: Benefits and Responsibility

The rapid proliferation of AI brings a complex mix of massive societal benefits alongside serious ethical challenges. [Carnegie Mellon Universit...](#)

+ Ask Gemini

Flash

Upgrade

The Breakthroughs

AI excels at processing scale. In medicine, it is used to analyze clinical trials and map proteins, accelerating drug discovery from decades to months. In daily operations, it eliminates tedious, repetitive tasks—allowing humans to focus on creative and high-level strategy. Furthermore, AI tools optimize energy grids, predict localized climate trends, and minimize human error in precision environments like manufacturing. [Oracle + 1](#)

The Challenges

Because AI learns from human-generated data, it inherently risks replicating human biases. If a model is trained on flawed data, it will produce flawed, discriminatory outcomes in sensitive areas like hiring or loan approvals. Additionally, the rapid rise of sophisticated generative tools has heightened concerns regarding misinformation, copyright infringement, and data privacy. [Google Cloud + 1](#)

The Path Forward: The ultimate success of AI does not depend on how smart the machines become, but on the framework humans build to govern them. The focus of the industry is heavily shifting toward Responsible AI—prioritizing transparency, explainability, and robust security data guardrails to ensure these powerful tools act safely as extensions of human capability. [IBM](#)

If you want to explore a specific angle of artificial intelligence further:

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environments like manufacturing. [Oracle + 1](#)

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If you want to explore a specific angle of artificial intelligence further:

Explain how Neural Networks actually learn

Explore the current timelines and debates around AGI

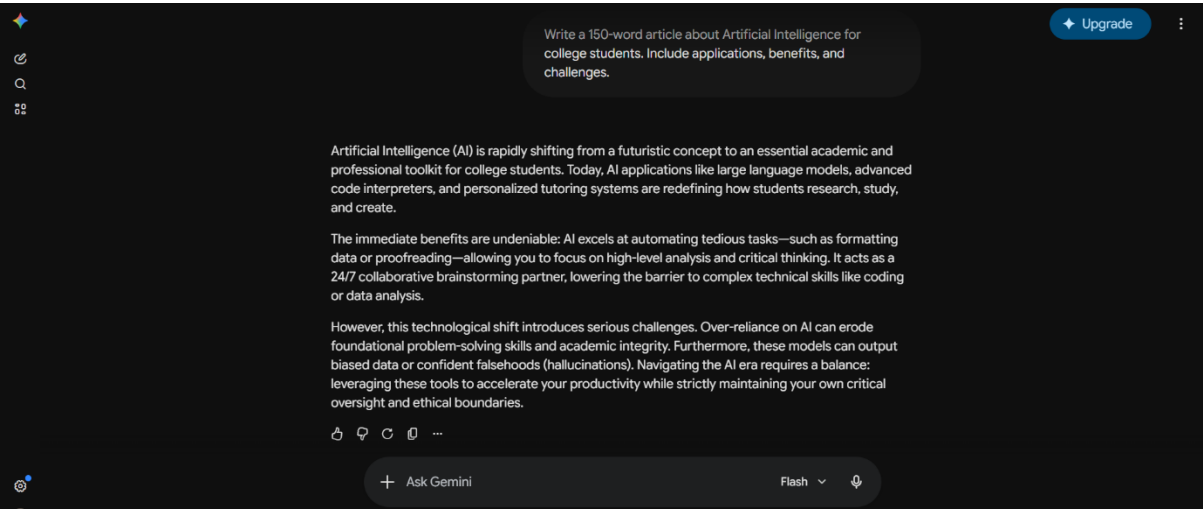
Analyze the impact of AI on the future job market

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Improved Prompt: Write a 150-word article about Artificial Intelligence for college students. Explain its applications, benefits, and challenges using clear and professional language.



Analysis:

The original prompt, "Write about AI," is vague and lacks specific instructions regarding the audience, length, format, and topics to be covered. As a result, the AI generated a broad and general response about Artificial Intelligence.

To improve the quality of the output, the prompt was rewritten with clear requirements, including:

- Target audience (college students)
- Content focus (applications, benefits, and challenges)
- Desired length (150 words)
- Professional tone

The improved prompt guided the AI to produce a more structured, informative, and relevant response. This demonstrates that prompt specificity significantly enhances the quality, accuracy, and usefulness of AI-generated content.

Key Observation

Aspect	Vague Prompt	Improved Prompt
Clarity	Low	High

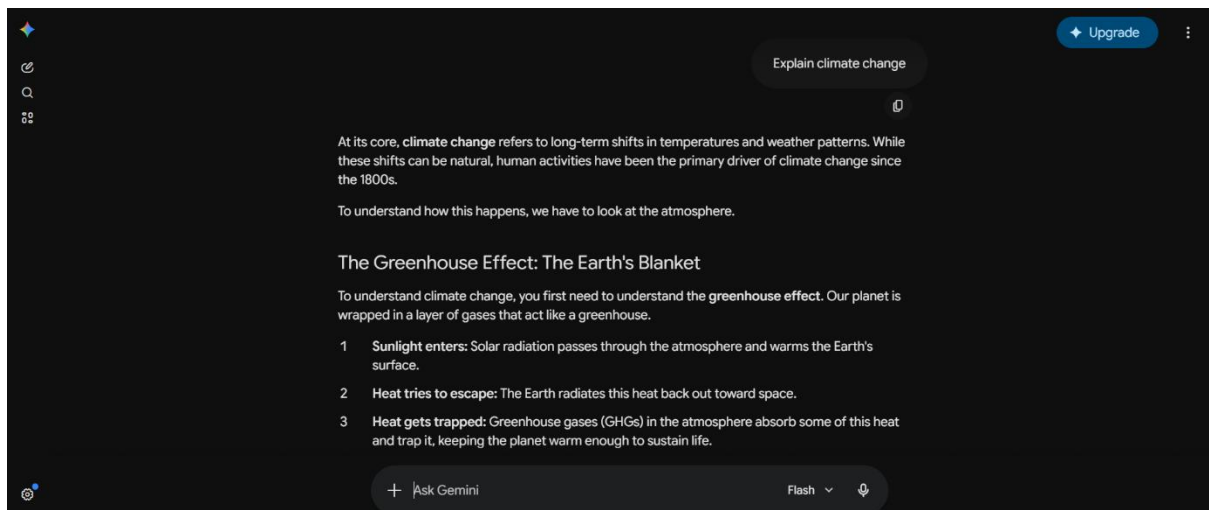
Aspect	Vague Prompt	Improved Prompt
Specificity	Low	High
Structure	General	Well-organized
Detail Level	Basic	Detailed
Relevance	Broad	Targeted

Conclusion:

This example shows that well-crafted prompts help AI understand the user's expectations more effectively, resulting in higher-quality outputs. Adding context, audience, format, and objectives leads to more accurate and useful responses.

2. Climate Change

Vague Prompt: Explain climate change.



The screenshot shows a Gemini AI chat window with a dark theme. The prompt "Explain climate change" is entered in the input field. The response is as follows:

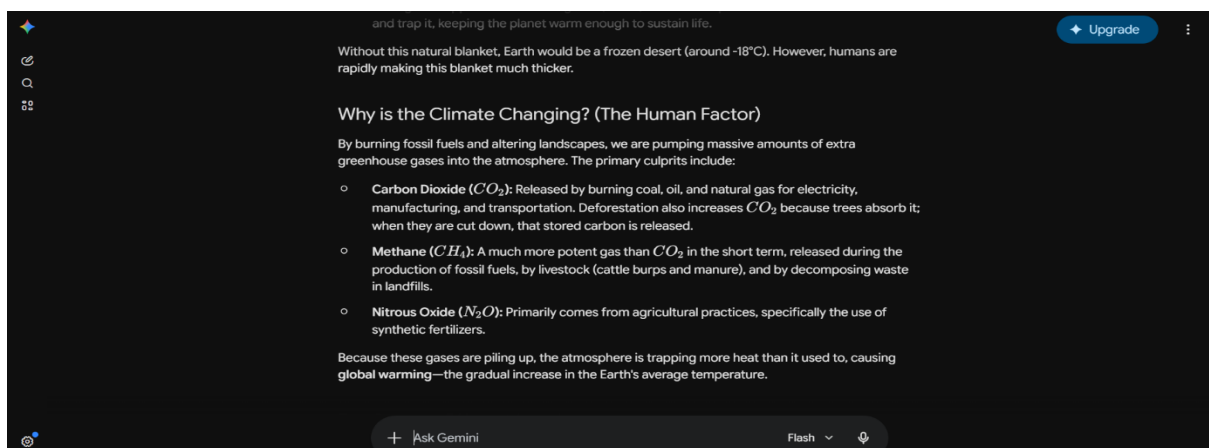
At its core, climate change refers to long-term shifts in temperatures and weather patterns. While these shifts can be natural, human activities have been the primary driver of climate change since the 1800s.

To understand how this happens, we have to look at the atmosphere.

The Greenhouse Effect: The Earth's Blanket

To understand climate change, you first need to understand the **greenhouse effect**. Our planet is wrapped in a layer of gases that act like a greenhouse.

- Sunlight enters:** Solar radiation passes through the atmosphere and warms the Earth's surface.
- Heat tries to escape:** The Earth radiates this heat back out toward space.
- Heat gets trapped:** Greenhouse gases (GHGs) in the atmosphere absorb some of this heat and trap it, keeping the planet warm enough to sustain life.



The screenshot shows the continuation of the Gemini AI chat window. The response continues from the previous one:

and trap it, keeping the planet warm enough to sustain life.

Without this natural blanket, Earth would be a frozen desert (around -18°C). However, humans are rapidly making this blanket much thicker.

Why is the Climate Changing? (The Human Factor)

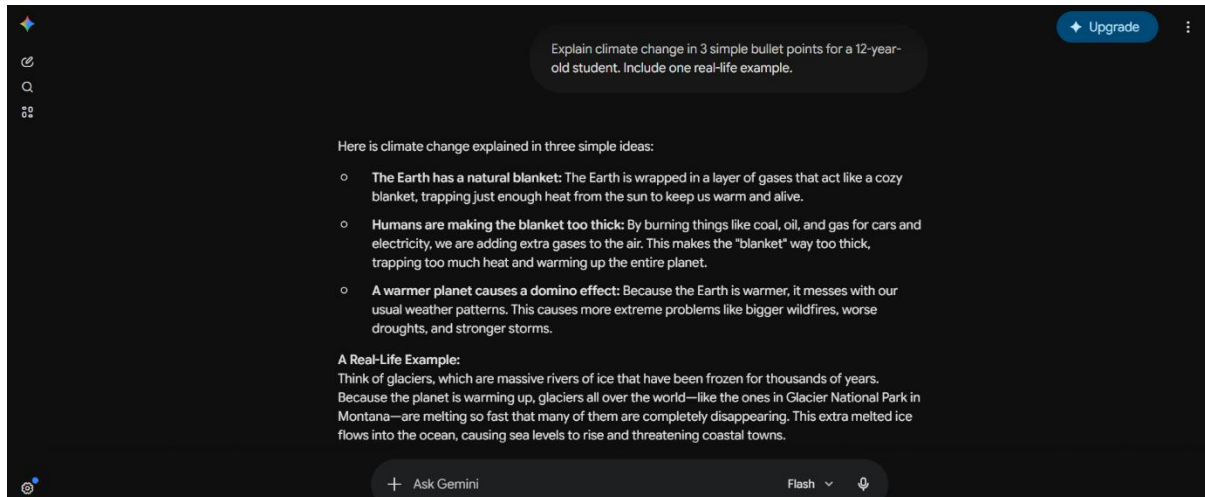
By burning fossil fuels and altering landscapes, we are pumping massive amounts of extra greenhouse gases into the atmosphere. The primary culprits include:

- Carbon Dioxide (CO_2):** Released by burning coal, oil, and natural gas for electricity, manufacturing, and transportation. Deforestation also increases CO_2 because trees absorb it; when they are cut down, that stored carbon is released.
- Methane (CH_4):** A much more potent gas than CO_2 in the short term, released during the production of fossil fuels, by livestock (cattle burps and manure), and by decomposing waste in landfills.
- Nitrous Oxide (N_2O):** Primarily comes from agricultural practices, specifically the use of synthetic fertilizers.

Because these gases are piling up, the atmosphere is trapping more heat than it used to, causing **global warming**—the gradual increase in the Earth's average temperature.

Improved Prompt:

Explain climate change in 3 simple bullet points for a 12-year-old student. Use easy language and include one real-life example.



Analysis

The original prompt, "Explain climate change," is too general and does not provide any information about the target audience, response format, or level of detail required. As a result, the AI generates a broad explanation that may not be suitable for all readers.

The improved prompt specifies:

- The target audience (12-year-old student)
- The response format (3 bullet points)
- The language style (easy language)
- The inclusion of a real-life example

These instructions help the AI generate a more structured, easy-to-understand, and engaging response. The output becomes more relevant to the intended audience and effectively communicates the concept of climate change.

Key Observation

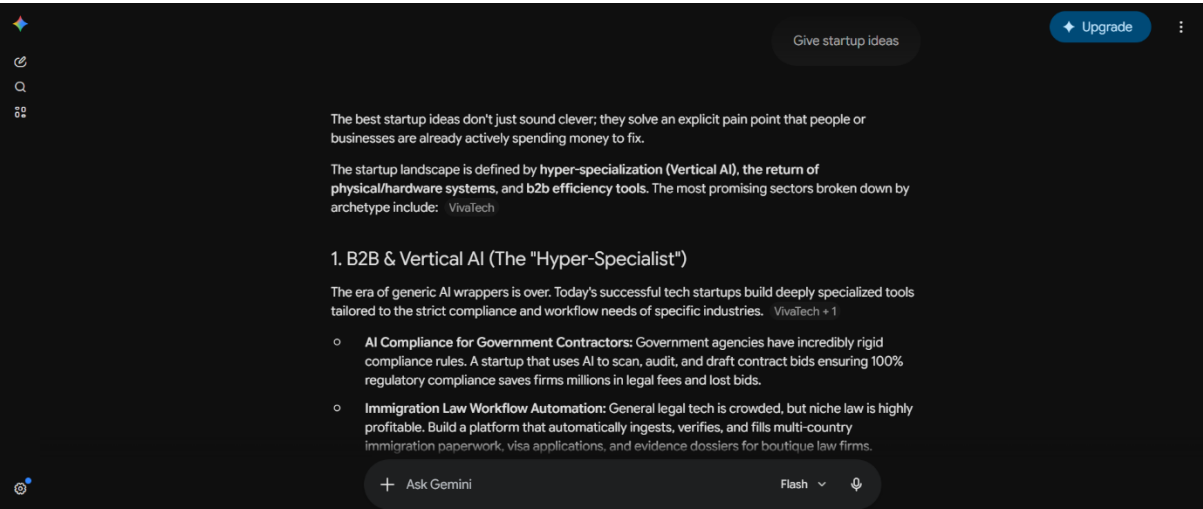
Aspect	Vague Prompt	Improved Prompt
Clarity	Low	High
Specificity	Low	High
Structure	General Paragraph	Bullet Points
Audience Focus	Not Specified	12-Year-Old Student
Detail Level	Basic	More Relevant
Understanding	Moderate	Easy to Understand

Conclusion:

This example demonstrates that adding clear instructions regarding audience, format, and content requirements significantly improves the quality and effectiveness of AI-generated responses.

3. Startup Ideas

Vague Prompt : Give startup ideas.



Wellows

Custom "Agent" Training & Orchestration: Companies want to deploy internal AI agents to handle HR onboarding, IT ticketing, or procurement. Start a specialized agency/consultancy that builds, audits, and fine-tunes bespoke internal corporate AI fleets.
U.S. Chamber of Commerce

Upgrade

2. HealthTech & Longevity (The "Hospital-at-Home")

With aging populations and strained medical infrastructure, tech that decentralizes healthcare out of hospitals and into homes is scaling rapidly.

- Elderly Care Coordination Platform:** A B2B2C marketplace or app that connects families, remote doctors, in-home nurses, and smart-home monitoring devices into a single dashboard to track an aging parent's health, medication adherence, and daily activities. [Wellows](#)
- Biometric-Driven Continuous Meal Planning:** Move beyond standard food logging. Build an API or app that syncs with Continuous Glucose Monitors (CGMs) and wearable data to dynamically order groceries or generate exact, biometrically optimized meal deliveries for users managing diabetes or longevity goals. [Wellows](#)
- Clinical AI Administrative Staff:** Doctors waste hours dictating notes and filing electronic health records (EHR). Startups that build passive, ambient audio listening tools that securely and accurately draft clinical notes natively into hospital legacy databases are winning massive enterprise contracts.

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3. Physical Tech & Logistics (The "Invisible Infrastructure")

After a decade heavily focused on pure software, massive value is migrating back into physical industries utilizing smart automation. [VivaTech](#)

Upgrade

- Autonomous Fleet Optimization Software:** With the rise of electric and semi-autonomous delivery fleets, logistics companies need predictive platforms. Build software that manages real-time battery degradation, route planning based on grid charging costs, and dynamic predictive maintenance.
- Warehouse "Physical AI" Integration:** Transitioning a traditional warehouse into a robotic one is incredibly buggy. Startups that act as the software layer connecting different hardware brands (e.g., matching automated forklifts with robotic item-pickers) are highly valuable.

4. Micro-SaaS & Creator Economy (The "Capital-Efficient")

If you want to launch with a small team and reach profitability fast without millions in venture capital funding, look at niche digital marketplaces.

- B2B Productivity Architecture Marketplace:** Modern companies run on complex Notion dashboards, Airtable databases, and Zapier/n8n automation pipelines. Building a premium marketplace specifically for enterprise-grade, plug-and-play workflow architectures yields immediate passive income.

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Upgrade

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- AI-Enhanced Video Repurposing Kits:** The creator economy is massive, but editing kills momentum. Build a highly specific tool that automatically takes long-form live streams, extracts the highest-engagement hooks using acoustic and visual analysis, and formats them instantly for multi-platform short-form video.

Evaluating Your Idea: Before choosing one, ask yourself: *Can I explain the core pain point in exactly one sentence? Do people already spend money trying to solve this poorly?*

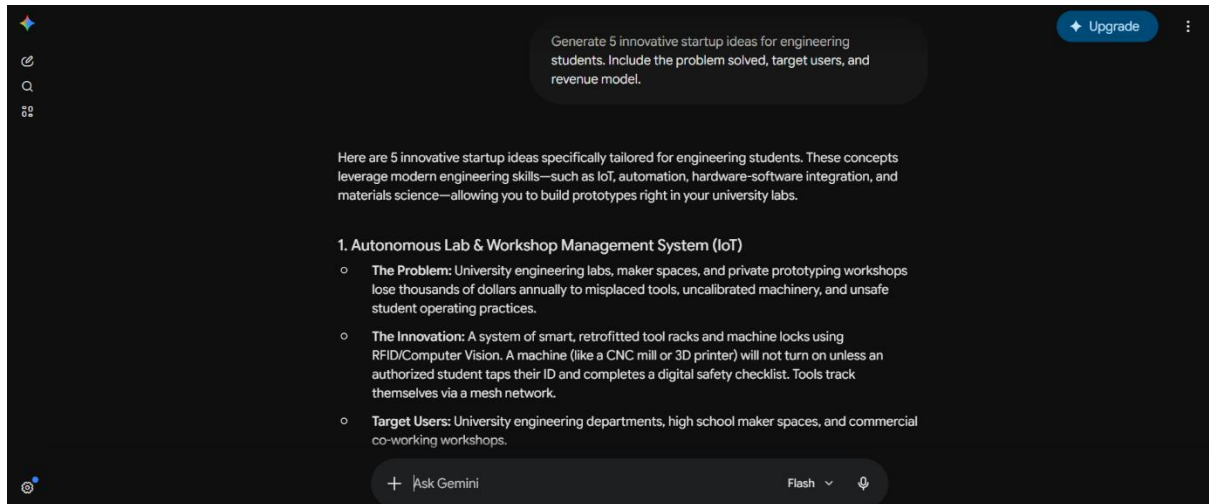
To give you a highly customized recommendation, what is your background, or what specific industry do you have the most experience in?

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Improved Prompt :

Generate 5 innovative startup ideas for engineering students. For each idea, include the problem solved, target users, and revenue model.



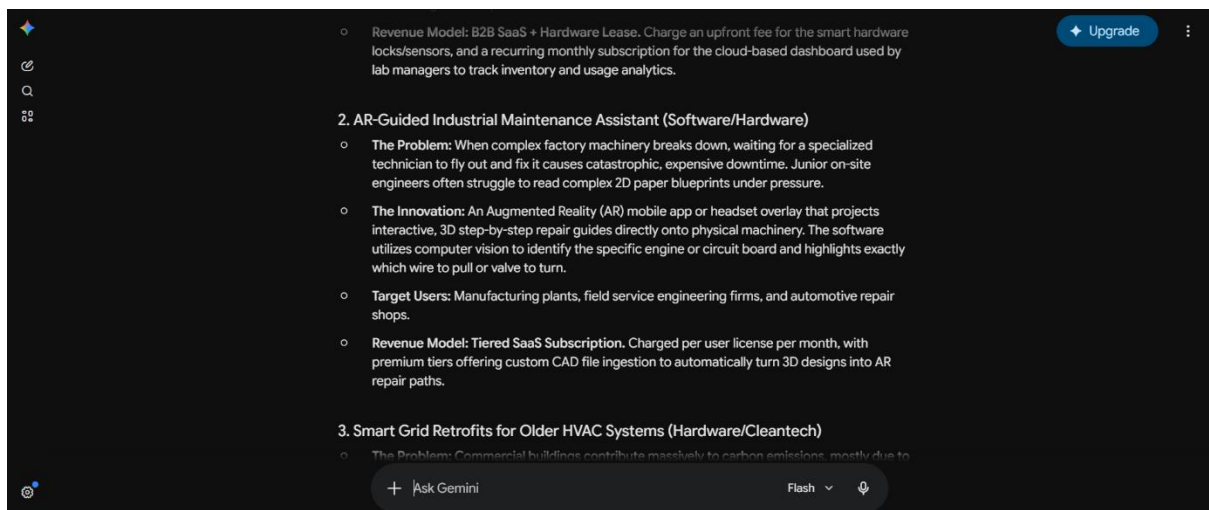
Generate 5 innovative startup ideas for engineering students. Include the problem solved, target users, and revenue model.

Here are 5 innovative startup ideas specifically tailored for engineering students. These concepts leverage modern engineering skills—such as IoT, automation, hardware-software integration, and materials science—allowing you to build prototypes right in your university labs.

1. Autonomous Lab & Workshop Management System (IoT)

- **The Problem:** University engineering labs, maker spaces, and private prototyping workshops lose thousands of dollars annually to misplaced tools, uncalibrated machinery, and unsafe student operating practices.
- **The Innovation:** A system of smart, retrofitted tool racks and machine locks using RFID/Computer Vision. A machine (like a CNC mill or 3D printer) will not turn on unless an authorized student taps their ID and completes a digital safety checklist. Tools track themselves via a mesh network.
- **Target Users:** University engineering departments, high school maker spaces, and commercial co-working workshops.

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- **Revenue Model:** B2B SaaS + Hardware Lease. Charge an upfront fee for the smart hardware locks/sensors, and a recurring monthly subscription for the cloud-based dashboard used by lab managers to track inventory and usage analytics.

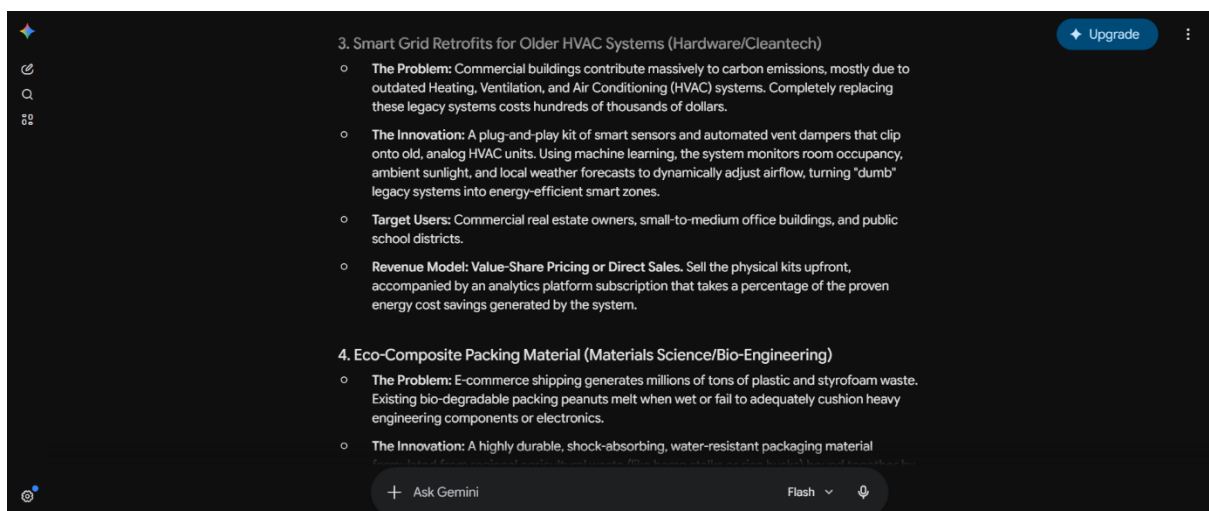
2. AR-Guided Industrial Maintenance Assistant (Software/Hardware)

- **The Problem:** When complex factory machinery breaks down, waiting for a specialized technician to fly out and fix it causes catastrophic, expensive downtime. Junior on-site engineers often struggle to read complex 2D paper blueprints under pressure.
- **The Innovation:** An Augmented Reality (AR) mobile app or headset overlay that projects interactive, 3D step-by-step repair guides directly onto physical machinery. The software utilizes computer vision to identify the specific engine or circuit board and highlights exactly which wire to pull or valve to turn.
- **Target Users:** Manufacturing plants, field service engineering firms, and automotive repair shops.
- **Revenue Model:** Tiered SaaS Subscription. Charged per user license per month, with premium tiers offering custom CAD file ingestion to automatically turn 3D designs into AR repair paths.

3. Smart Grid Retrofits for Older HVAC Systems (Hardware/Cleantech)

- **The Problem:** Commercial buildings contribute massively to carbon emissions, mostly due to outdated Heating, Ventilation, and Air Conditioning (HVAC) systems. Completely replacing these legacy systems costs hundreds of thousands of dollars.

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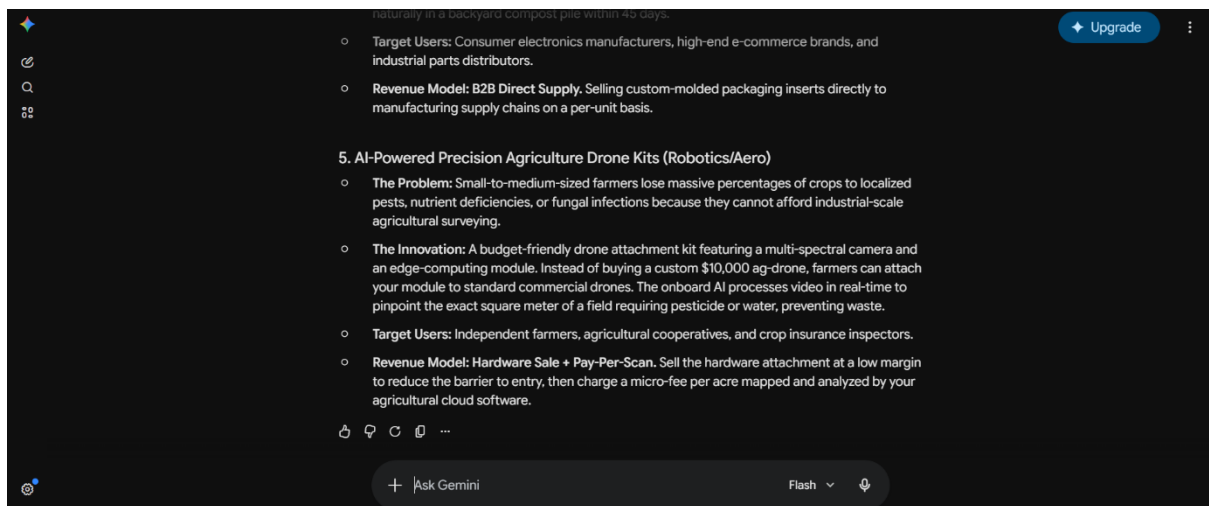
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- **The Problem:** Commercial buildings contribute massively to carbon emissions, mostly due to outdated Heating, Ventilation, and Air Conditioning (HVAC) systems. Completely replacing these legacy systems costs hundreds of thousands of dollars.
- **The Innovation:** A plug-and-play kit of smart sensors and automated vent dampers that clip onto old, analog HVAC units. Using machine learning, the system monitors room occupancy, ambient sunlight, and local weather forecasts to dynamically adjust airflow, turning "dumb" legacy systems into energy-efficient smart zones.
- **Target Users:** Commercial real estate owners, small-to-medium office buildings, and public school districts.
- **Revenue Model:** Value-Share Pricing or Direct Sales. Sell the physical kits upfront, accompanied by an analytics platform subscription that takes a percentage of the proven energy cost savings generated by the system.

4. Eco-Composite Packing Material (Materials Science/Bio-Engineering)

- **The Problem:** E-commerce shipping generates millions of tons of plastic and styrofoam waste. Existing bio-degradable packing peanuts melt when wet or fail to adequately cushion heavy engineering components or electronics.
- **The Innovation:** A highly durable, shock-absorbing, water-resistant packaging material

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Analysis

The original prompt, "Give startup ideas," is very broad and does not specify the target audience or the type of information required. As a result, the AI generates generic startup ideas with minimal detail.

The improved prompt clearly defines:

- The target audience (engineering students)
- The number of ideas required (5 ideas)
- The information to include (problem solved, target users, and revenue model)

These instructions help the AI generate more relevant, structured, and detailed startup ideas. The response becomes more useful for evaluating business opportunities and understanding how each idea could be implemented.

Key Observation

Aspect	Vague Prompt	Improved Prompt
Clarity	Low	High
Specificity	Low	High
Structure	Simple List	Detailed Format

Aspect	Vague Prompt	Improved Prompt
Audience Focus	Not Specified	Engineering Students
Creativity	Basic	More Innovative
Detail Level	Low	High

Conclusion :

This example demonstrates that adding specific requirements and context to a prompt results in more detailed, organized, and valuable AI-generated responses. Clear prompts help the AI better understand the user's expectations and generate higher-quality outputs.

Final Conclusion:

This task demonstrated that the quality of AI - generated responses depends heavily on the quality of the prompts provided. By adding clear instructions, defining the target audience, specifying the format, and including relevant details, AI can generate more accurate, structured, and useful outputs. Effective prompt engineering is therefore an essential skill for obtaining high-quality results from AI systems.